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Roll No

CE-703(D)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Building Services

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain how air is conditioned and distributed to the various zones from a AHU through duct systems. 7
b) Write short notes of following (any two): 7
 - i) Rain water harvesting
 - ii) Types Car parking systems
 - iii) Various security measures in a office building
2. a) Differentiate the following: 7
 - i) Non-scouring and self cleansing velocities
 - ii) Dry and water carriage system of sewerageb) Differentiate the following: 7
 - i) Sewer, sewage and sewerage
 - ii) Waste branch and soil branch of sewerage
3. a) Draw the following (any two): 7
 - i) Bell and spigot joint
 - ii) Check valve
 - iii) Flanged jointb) Draw the following (any two): 7
 - i) Sedimentation tank in treatment plant
 - ii) Drop manhole
 - iii) Water closet

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4. a) With the help of a line diagram explain the construction and functions of automatic sprinkler systems and explain its types. 7
b) Write short notes on Advantages of Automatic fighting systems over other fire fighting systems. 7
5. a) Describe the parts and functioning of a simple elevator, explain through neat plan and section. Discuss the provision of arrangement in lift lobby. 7
b) Explain the difference between Evaporative Cooler and Cooling Tower. 7
6. a) What is the difference between unitary and central air condition system and what are their various types? Explain with neat sketches. 7
b) What principles are being involved in the working of a septic tank and draw the cross section. 7
7. Design a public toilet for an office building as mentioned below: 14
 - i) W. C. (Indian) 3 Nos.
 - ii) Urinal 3 Nos.
 - iii) Wash Basin 2 Nos.Drawing requirements are as follows:-
 - a) Layout plan showing the detail of various sanitary appliances with their connectivity to the inspection chamber to the private sewer line.
 - b) Cold water supply to the required places in the same plan.
8. a) Explain with sketches various types of lift controlling systems. 7
b) With the help of sketches explain the following terms: 7
 - i) Noise rating curve
 - ii) Reverberation time

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